

CHOOSE THE CORRECT ANSWER

(30 POINTS)

الاجابة تكون في الجدول الموجود باسفل صفحة الغلاف

1. CIL02

(Variation of parameters method)

For the differential equation

$$x^2 y'' - 4xy' + 6y = x^2 \ln x, \quad x > 0$$

if the particular solution ;

$$y_p = (x^2)u_1 + (x^3)u_2 \quad \text{then } u_1' =$$

A. $-\frac{\ln x}{x^2}$

B. $\frac{\ln x}{x}$

☒ C. $-\frac{\ln x}{x}$

D. $\frac{\ln x}{x^2}$

2. CIL01

The integrating factor that makes the differential equation;

$$(xy)dx + (5+y)dy = 0, \quad x > 0, y > 0$$

to be exact, is:

☒ A. $\mu(y) = y^{-1}$

B. $\mu(x) = x^{-2}$

C. $\mu(x) = x^{-1}$

D. $\mu(y) = y^{-2}$

3. CIL01

The general solution of the following differential equation;

$$y' + \frac{y}{x} = 2 \sin x^2, \quad x > 0$$

is:

☒ A. $y = \frac{1}{x} (-\cos x^2 + c)$

C. $y = \frac{1}{x} (\cos x^2 + c)$

B. $y = \frac{1}{x} \cos x^2 + c$

D. $y = -\frac{1}{x} \cos x^2 + c$

4. CIL01

The assumption that converts the differential equation ;

$$y' + 2xy = \frac{x}{y}, \quad x > 0, y > 0$$

to be linear is:

A. $u = y^3$

B. $u = y^{-1}$

☒ C. $u = y^2$

D. $u = y^{-2}$

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5. CILO5

The general solution of the following differential equation;

$$\frac{dy}{dx} = \sqrt{xy} \quad \text{is:}$$

A. $2\sqrt{y} = \sqrt{x^3} + c$

B. $\sqrt{y} = \sqrt{x^3} + c$

☒ C. $3\sqrt{y} = \sqrt{x^3} + c$

D. $\sqrt{y} = 3\sqrt{x^3} + c$

6. CILO8

If the linear differential equation;

$$y'' + y' - 4 = f(x) \quad \text{is homogeneous then } f(x) =$$

A. 2

B. 0

C. 4

☒ D. -4

7. CILO5

The general solution of the following differential equation;

$$\frac{dy}{dx} = -1 + \sqrt{4 - (x+y)^2} \quad \text{is:}$$

A. $y = -x - 2\sin(x+c)$

B. $y = -x - 4\sin(x+c)$

C. $y = -x + 4\sin(x+c)$

☒ D. $y = -x + 2\sin(x+c)$

8. CILO1

Which of the following differential equation, is linear ?

A. $(y')^2 + 2xy = x$

☒ C. $y' + 2xy = x^2$

B. $yy' + 2xy = x^2$

D. $y' + 2xy = \sqrt{y}$

9. CILO2

(Method of undetermined coefficients)

Write the suitable assumption for the particular solution, in order to solve the following differential equation;

$$y''' - y'' = x + e^x$$

☒ A. $y_p = Ax^3 + Bx^2 + Cxe^x$

C. $y_p = Ax^3 + Bx^2 + Ce^x$

B. $y_p = Ax + B + Ce^x$

D. $y_p = Ax^2 + Bx + Ce^x$

10. CILO6

Write the non-homogeneous linear differential equation, whose general solution is

$$y = c_1 e^{10x} + c_2 e^{-10x} + \sin 2x \quad ?$$

A. $y'' + 100y = -104 \sin 2x$

C. $y'' - 100y = -96 \cos 2x$

☒ B. $y'' - 100y = -104 \sin 2x$

D. $y'' + 100y = -96 \cos 2x$

11. CULO1

The degree and the order of the differential equation;

$$\left(y^{(4)}\right)^3 + (y'')^7 = yy''' \text{ are:}$$

- A. 4 and 3 respectively
B. 7 and 4 respectively

- ☒ C. 3 and 4 respectively
D. 4 and 7 respectively

12. CULO8If the characteristic equation of the homogeneous Cauchy-Euler differential equation, is $m^3 - 1 = 0$ then the differential equation is:

- A. $x^3 y''' + y = 0$
B. $x^3 y''' - y = 0$

- ☒ C. $x^3 y''' + 3x^2 y'' + xy' - y = 0$
D. $x^3 y''' + 3x^2 y'' + xy' + y = 0$

13. CULO2

Solve the (IVP);

$$x^2 y'' - 2xy' + 2y = 0, \quad y(1) = 2, \quad y'(1) = 4$$

- A. $y = 2x^3 - x^2 + 1$
B. $y = x^4 + 1$

- ☒ C. $y = 2x^2$
D. $y = x^2 + 4x - 3$

14. CULO2

The general solution of the differential equation;

$$y''' - 4y' = 0$$

is:

A. $y = c_1 + c_2 e^{-2x} + c_3 x e^{-2x}$

B. $y = c_1 + c_2 e^{2x} + c_3 x e^{2x}$

☒ C. $y = c_1 + c_2 e^{2x} + c_3 e^{-2x}$

D. $y = c_1 e^{2x} + c_2 e^{-2x} + c_3 x e^{-2x}$

15. CULO6Write the homogeneous linear differential equation with constant coefficients, if the roots of its characteristic equation are $3 \pm \sqrt{2}i$?

- A. $y'' - 6y' + 11 = 0$
B. $y'' - 6y' - 11 = 0$

- ☒ C. $y'' - 6y' + 11y = 0$
D. $y'' + 6y' + 11y = 0$

Q1) One of the following differential equations is of order 3 and degree 2

a) $y'' - y' = 0$

~~b) $e^x(y')^3 + x^2(y'')^3 = y$~~

✓ c) $2x\sqrt{y''} - 4(y')^2 + y' = 0$

~~d) $(y''')^3 + (y'')^2 + y' = 0$~~

Q2) The general solution of the D.E. $\frac{dy}{dx} = \frac{xy + 2x - y - 2}{xy - 2x + y - 2}$ is

a) $\ln(y+2)^4 = \ln(x-1)^2 + c$

(b) $y - 4\ln|y+2| = x - 2\ln|x+1| + c$

c) $\frac{y^2}{2} - 4y = \frac{x^2}{2} + 2x + c$

d) $y - 4\ln|y+2| = x + 2\ln|x-1| + c$

Q3) One of the following differential equations is not homogeneous

a) $xyy' = (x+y)^2 e^{\frac{x}{y}}$

(b) $(x^2 e^{y^2} + xy)dx + (y^2 - 2xy)dy = 0$

c) $yy' = x - \sqrt{x^2 - y^2}$

d) $2xyy' = 2x^2 - 3y^2$

Q4) The general solution of the D.E. $y' = (x+y)^2$ is

a) $\frac{y^3}{3} - \frac{y^2}{2} + y = \frac{x^3}{3} + 2x + c$

b) $\frac{y^2}{2} = \frac{x^2}{2} + 2x + c$

c) $y = \frac{e^{2x}}{2} + c$

(d) $\tan^{-1}(x+y) = x + c$

Q5) If the D.E. $g(x,y)dx + (e^{xy} + \cos(2x+3y))dy = 0$ is Exact, then $\frac{\partial g}{\partial y} =$

a) $xe^{xy} - 3\sin(2x+3y)$

✓ (b) $ye^{xy} - 2\sin(2x+3y)$

c) $ye^{xy} - 3\sin(2x+3y)$

d) $xe^{xy} - 2\sin(2x+3y)$

Q6) If $F(x,y) = c$ is the general solution of the D.E.

$(2xe^y + y^2)dx + (x^2e^y + 2xy)dy = 0$, then $\frac{\partial F}{\partial y} =$

a) $2e^y + 2y$

b) $x^2e^y + 2x$

(c) $x^2e^y + 2xy$

d) $2xe^y + y^2$

Q7) The integrating factor of the linear D.E. $\sin(x)y' + 2\cos(x)y + \sin(x)y = e^{-x}\cos x$ is

a) $e^{\sin^2 x}$

b) $e^{\cot x}$

c) $\sin^2(x)e^{2x}$

(d) $\sin^2(x)e^x$

Q8) Any first order homogeneous D.E. is converted to

(a) Separable D.E.

b) Linear D.E.

c) Exact D.E.

d) Bernoulli's D.E.

Q9) By converting the Bernoulli's D.E. $y^2 y' + 2xy^3 = \sin x$ to linear D.E. we get

a) $u' + 9xu = 3\sin x$ **(b)** $u' + 6xu = 3\sin x$ ~~c)~~ $u' + 2xu = 2\sin x$ d) $u' - 2xu = \sin x$

Q10) If $y_1 = x$ is the solution of the D.E. $x^2 y'' - x(x+2)y' + (x+2)y = 0$, then $y_2 =$

a) xe^x **(b)** $x^2 e^x$ c) $\frac{e^x}{x}$ d) $x^3 e^x$

Q11) $y = \frac{c_1 \sin \ln x^3 + c_2 \cos \ln x^3}{x}$ is the general solution of the D.E.

~~a)~~ $y'' + 2y' + 10y = 0$ ~~b)~~ $y'' + 3y' + 10y = 0$
(c) $x^2 y'' + 2xy' + 10y = 0$ d) $x^2 y'' + 3xy' + 10y = 0$

Q12) $y = e^x (c_1 + c_2 \sin 3x + c_3 \cos 3x)$ is the general solution of the D.E.

a) $y''' + 9y' = 0$ b) $y''' - 2y'' + 6y' - 9y = 0$
c) $y''' - 2y'' + 8y' - 10y = 0$ **(d)** $y''' - 3y'' + 12y' - 10y = 0$

Q13) $y = c_1 x + c_2 x \ln x + c_3 x (\ln x)^2$ is the general solution of the D.E.

a) $x^3 y''' + xy' - y = 0$ b) $x^3 y''' + 2x^2 y'' + 2xy' - y = 0$
c) $x^3 y''' + 2x^2 y'' + xy' - y = 0$ d) $x^3 y''' + x^2 y'' + 2xy' - y = 0$

Q14) The solution of the I.V.P. $y'' - 2y' - 3y = 0$, $y(0) = 4$, $y'(0) = 0$ is

(a) $y = 3e^{-x} + e^{3x}$ ~~b)~~ $y = 4e^{-x} - e^{3x}$ ~~c)~~ $y = 3e^x + e^{-3x}$ ~~d)~~ $y = 6e^{-x} + 2e^{3x}$

Q15) The largest interval at which the I.V.P

$\ln(x+3)y'' + xy' + (\sec x)y = \sqrt[3]{x}$, $y(2) = 1$, $y'(2) = 0$ has unique solution is :

(where $\pi \approx 3.14$, $\frac{\pi}{2} \approx 1.57$, $\frac{3\pi}{2} \approx 4.71$)

a) $(-3, \infty)$ b) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ **(c)** $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ d) $\left(\frac{\pi}{2}, \infty\right)$

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1. CILOI

The order and the degree of the differential equation;

$$\left(y^{(4)}\right)^3 + (y'')^7 = yy''' \text{ are:}$$

A) 3 and 4 respectively

☒ B) 4 and 3 respectively

C) 7 and 4 respectively

D) 4 and 7 respectively

2. CILOI

Which of the following differential equation, is linear ?

~~A~~ $yy' + 2xy = x^2$

~~C~~ $y' + 2xy = x^2 \sqrt{y}$

☒ B) $y' + 2xy = \sqrt{x}$

~~D~~ $(y')^2 + 2xy = x$

3. CILOI $y' + p(x)y = g(x)$

The integrating factor that makes the differential equation;

$$(2xy^2)dx + (5+y)dy = 0, x > 0, y > 0$$

to be exact, is:

☒ A) $\mu(y) = y^{-2}$

B) $\mu(y) = y^{-1}$

C) $\mu(x) = x^{-2}$

D) $\mu(x) = x^{-1}$

4. CILOI

The general solution of the following differential equation;

$$y' + \frac{y}{x} = 2 \cos x^2, x > 0$$

is:

~~A~~ $y = \frac{1}{x} \sin x^2 + c$

☒ C) $y = -\frac{1}{x} \sin x^2 + c$

B) $y = -\frac{1}{x} (\sin x^2 + c)$

~~D~~ $y = \frac{1}{x} (\sin x^2 + c)$

5. CILOI

The assumption that converts the differential equation ;

$$y' + 2xy = \frac{x}{y^2}, x > 0, y > 0$$

to be linear is:

☒ A) $u = y^3$

B) $u = y^{-2}$

C) $u = y^2$

D) $u = y^{-3}$

6. Q102

The general solution of the differential equation;

$$y''' - 9y' = 0$$

is:

A) $y = c_1 + c_2 e^{3x} + c_3 x e^{3x}$

B) $y = c_1 + c_2 e^{-3x} + c_3 x e^{-3x}$

C) $y = c_1 e^{3x} + c_2 e^{-3x} + c_3 x e^{-3x}$

D) $y = c_1 + c_2 e^{3x} + c_3 e^{-3x}$

7. Q103

Solve the (IVP);

$$x^2 y'' - 2xy' + 2y = 0, \quad y(1) = 3, \quad y'(1) = 6$$

A) $y = x^6 + 1$

C) $y = 4x + x^2 - 2$

B) $y = 3x^2$

D) $y = x^2 + x^4 + 1$

8. Q104

(Method of undetermined coefficients)

Write the suitable assumption for the particular solution, in order to solve the following differential equation;

$$y''' + y'' = x + e^{-x}$$

A) $y_p = Ax + B + Ce^{-x}$

C) $y_p = Ax^3 + Bx^2 + Ce^{-x}$

B) $y_p = Ax^3 + Bx^2 + Cxe^{-x}$

D) $y_p = Ax^2 + Bx + Ce^{-x}$

9. Q105

(Variation of parameters method)

For the differential equation

$$x^2 y'' - 4xy' + 6y = x^2 \ln x, \quad x > 0$$

if the particular solution ;

$$y_p = (x^2)u_1 + (x^3)u_2 \quad \text{then } u_2' =$$

A) $\frac{\ln x}{x^2}$

B) $-\frac{\ln x}{x}$

C) $-\frac{\ln x}{x^2}$

D) $\frac{\ln x}{x}$

10. Q106

The general solution of the following differential equation;

$$\frac{dy}{dx} = -1 + \sqrt{16 - (x+y)^2} \quad \text{is:}$$

A) $y = -x - 2 \sin(x+c)$

C) $y = -x + 4 \sin(x+c)$

B) $y = -x + 2 \sin(x+c)$

D) $y = -x - 4 \sin(x+c)$

11. CU05

The general solution of the following differential equation;

$$\frac{dy}{dx} = \sqrt[3]{xy} \quad \text{is:}$$

A) $3 \sqrt[3]{y^2} = \sqrt[3]{x^4} + c$

B) $\sqrt[3]{y^2} = 3 \sqrt[3]{x^4} + c$

☒ C) $2 \sqrt[3]{y^2} = \sqrt[3]{x^4} + c$

D) $\sqrt[3]{y^2} = 2 \sqrt[3]{x^4} + c$

12. CU06Write the homogeneous linear differential equation with constant coefficients, if the roots of its characteristic equation are $2 \pm \sqrt{3}i$?

A) $y'' + 4y' + 7y = 0$

C) $y'' - 4y' + 7 = 0$

☒ B) $y'' - 4y' + 7y = 0$

D) $y'' - 4y' - 7 = 0$

13. CU06

Write the non-homogeneous linear differential equation, whose general solution is

$$y = c_1 e^{10x} + c_2 e^{-10x} + \cos 2x \quad ?$$

A) $y'' - 100y = -104 \cos 2x$

C) $y'' - 100y = 104 \cos 2x$

☒ B) $y'' + 100y = -104 \cos 2x$

D) $y'' + 100y = 104 \cos 2x$

14. CU08

If the linear differential equation;

$$y'' + y' - 6 = f(x) \quad \text{is homogeneous then } f(x) =$$

A) 0

B) 6

C) -2

☒ D) -6

15. CU08If the characteristic equation of the homogeneous Cauchy-Euler differential equation, is $m^3 + 1 = 0$ then the differential equation is:

A) $x^3 y''' + 3x^2 y'' + xy' - y = 0$

C) $x^3 y''' - y = 0$

☒ B) $x^3 y''' + 3x^2 y'' + xy' + y = 0$

D) $x^3 y''' + y = 0$

4

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	B	B	A	D	A	D	B	B	A	C	C	B	A	D	B
CILO	CILO1	CILO1	CILO1	CILO1	CILO1	CILO2	CILO2	CILO2	CILO2	CILO3	CILO3	CILO6	CILO6	CILO8	CILO8